

MYELOID NEOPLASIA

Chromosome 5q deletion drives evolution of aneuploidy in myeloid neoplasms with complex karyotype

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KEY POINTS

- **del(5q) drives progressive chromosome evolution in *TP53*-mutant preleukemic HSPCs.**
- **Aneuploid HSPCs exhibit a shared gene expression signature that confers distinct therapeutic vulnerabilities in AML-CK.**

Clonal acquisition of multiple chromosomal abnormalities in hematopoietic stem and progenitor cells (HSPCs) is a hallmark of high-risk acute myeloid leukemia with complex karyotype (AML-CK). AML-CK is associated with *TP53* mutations and chromosome 5q deletions [del(5q)]; however, the drivers and clonal trajectories of aneuploid evolution in HSPCs remain unknown. We developed a patient-derived induced pluripotent stem cell (iPSC) model in which preleukemic HSPCs clonally evolve to distinct, highly aneuploid states after transient mitotic inhibition. By tracking chromosome evolution at single-cell resolution, we show that *TP53*-mutant HSPCs with del(5q), but not *TP53* mutation alone, evolve complex chromosomal changes. Clonal evolution was marked by the stepwise acquisition of numerical and structural chromosome changes seen in patients with AML-CK, with individual abnormalities conferring a fitness advantage. iPSC-derived aneuploid HSPCs and primary samples from patients with AML-CK exhibited a conserved gene expression signature marked by upregulation of *PTEN*, cohesins, and the antiapoptotic

factor *BCL2*, indicative of a shared aneuploid cell state in HSPCs. The clinical *BCL2* inhibitor venetoclax eradicated *BCL2*-dependent aneuploid clones, with resistant clones undergoing a lineage switch to upregulate alternative *BCL2* factors. In summary, we demonstrate that mutant *TP53* and del(5q) drive chromosome evolution marked by stepwise acquisition of individual abnormalities. Moreover, aneuploid HSPCs exhibit a shared gene expression state that confers unique, targetable therapeutic vulnerabilities in AML-CK.

Introduction

Aneuploidy involves changes in chromosome number or structure¹⁻³ and is pervasive in cancers.⁴ Aneuploidy is a hallmark of high-risk acute myeloid leukemia (AML) with complex karyotype (AML-CK), defined as ≥ 3 chromosomal abnormalities.⁵ AML-CK is almost universally associated with mutations

in *TP53*, very poor outcomes, and lower response to current antileukemic therapies.⁶⁻⁸ Most patients with AML-CK present with widespread aneuploidy at diagnosis, making it challenging to reconstruct clonal trajectories of early disease. It remains unknown how hematopoietic stem and progenitor cells (HSPCs) with aneuploidy first arise and evolve to ultimately give rise to these aggressive leukemias.

Although relatively infrequent in de novo AML, complex karyotype with *TP53* mutations is enriched in therapy-related myeloid neoplasms, implicating cytotoxic therapy in inducing chromosome instability or creating selective pressure for mutant clones.⁹ *TP53*-mutant clones are detectable at low frequency years before a diagnosis of therapy-related myeloid neoplasm and, in many cases, are antecedent to chemotherapy.^{10,11} *TP53* mutations are found in age-related clonal hematopoiesis, which is associated with an increased risk of myeloid malignancy.^{12,13} These observations broadly suggest that *TP53* mutations are early events in AML-CK pathogenesis and that clonal progression to AML-CK requires the occurrence of additional genetic events often in the context of cytotoxic therapy. Despite the advent of improved single-cell copy number detection methods, detection of early clones in this disease remains rare.¹⁴ Consequently, little is known about the genetic drivers and trajectories of AML-CK evolution from preleukemic HSPCs.

Specific chromosomal abnormalities are recurrent in certain cancers, suggesting that they affect genes that contribute to cancer fitness and progression.^{15,16} Most patients with AML-CK harbor recurrent deletions on chromosomes 5q [del(5q)], 7q, and/or 17p (encoding *TP53*), with del(5q) being the most common change found in over 80% of AML-CK cases, suggesting that this region encodes critical genes for disease progression.^{7,17-19} Classical studies have established the contribution of individual 5q genes to leukemogenesis.¹⁹⁻²³ However, it remains unknown if del(5q) acts as a driver of AML-CK progression, owing to a lack of experimental model systems in which normal HSPCs evolve to complex karyotypes. Understanding the role of del(5q) as a driver in AML-CK could nominate mechanisms and molecular dependencies because of haploinsufficiency of 5q-encoded genes. More broadly, aneuploid states may confer specific molecular dependencies due to cellular stresses on proteostasis, replication, and mitosis,²⁴ which could be therapeutically harnessed for incurable *TP53*-mutated leukemias.²⁵

Here, we report an induced pluripotent stem cell (iPSC) model in which patient-derived *TP53*-mutant preleukemic HSPCs evolve to distinct, highly aneuploid states after mitotic mis-segregation. We show that HSPCs with *TP53* mutation and del(5q) undergo progressive aneuploid evolution resembling that seen in patients with AML-CK. By tracking the trajectories of chromosome evolution at the single-cell level, we show that aneuploid evolution is characterized by stepwise acquisition of abnormalities that converge on a shared gene expression signature in AML-CK, revealing unique molecular dependencies associated with acquisition of aneuploidy in hematopoietic stem cells (HSCs) that can be therapeutically targeted in *TP53*-mutated leukemias.

Materials and methods

MDS/AML iPSC-derived HSPC culture

TP53^{mut} and *TP53*^{mut} del(5q) iPSC lines were previously derived from bone marrow cells of a female patient with myelodysplastic syndrome (MDS)/AML-CK.^{20,26} Primary AML samples were obtained from the Fred Hutchinson Cancer Center Co-Operative Center for Excellence in Hematology

core or the Mayo Clinic Biorepository for Acute Leukemia Research after institutional review board approval. Written informed consent was obtained from all participants. iPSC lines were differentiated into HSPCs and established as expandable iPSC-HSPCs by the introduction of 5 transcription factors.^{20,26} iPSC-HSPCs were cultured in StemSpan SFEM II (Stemcell Technologies) plus 50 ng/mL stem cell factor, 50 ng/mL FMS-like tyrosine kinase 3 ligand, 50 ng/mL thrombopoietin, 50 ng/mL interleukin-6, 10 ng/mL interleukin-3 (PeproTech), and doxycycline (2 µg/mL; Sigma).²⁷ Cells were maintained at 0.5×10^6 to 1×10^6 cells per mL, and media were changed every 3 to 4 days. For evolution to aneuploidy, iPSC-HSPC cell lines were treated with 1 µM of the monopolar spindle 1 kinase (MPS1) inhibitor NMS-P715 (Sigma) in dimethyl sulfoxide for 24 hours, followed by a complete media change. Cultures were maintained at 0.5×10^6 to 1×10^6 cells per mL and cultured for 60 to 70 days or as specified. Complete details are provided in supplemental Methods, available on the *Blood* website.

Single-cell RNA sequencing and CNI analysis

Patient samples were thawed by dropwise addition of Iscove modified Dulbecco medium plus 1% bovine serum albumin, washed, and then purified using the CD34⁺ MicroBead Kit (Miltenyi), followed by the Dead Cell Removal Kit (Miltenyi). iPSC-HSPC cell lines were thawed, and dead cells were removed using the Dead Cell Removal kit. Single-cell sequencing was performed using the 10x Chromium Next GEM Single Cell 3' Reagents kit version 3.1 on a Chromium Controller. Libraries were sequenced using Illumina NGS NextSeq/NovaSeq 150/100 PE cycle kits. After alignment to hg38 using the 10x Genomics Cell Ranger count tool, the Seurat R Package²⁸ version 4.1.3 was used to perform integration when multiple runs were analyzed together, with mitochondrial gene regression (SCTransform function), dimensional reduction, uniform manifold approximation and projection generation, and clustering. Copy number inference (CNI) was performed using Numbat.²⁹ Complete details are provided in supplemental Methods.

Results

Experimental model of aneuploid evolution in preleukemic HSPCs

We previously reprogrammed bone marrow cells from a patient with Li-Fraumeni syndrome who developed AML-CK and isolated 2 isogenic preleukemic iPSC lines: an ancestral line with a heterozygous *TP53* R209fs mutation (hereafter, *TP53*^{mut}) and a derived line with *TP53* R209fs and an isolated del(5)(q22q31) (hereafter, *TP53*^{mut} del(5q)).^{20,26} Reprogramming of these clones suggested that *TP53*-mutant preleukemic clones first acquired isolated del(5q), followed by other complex abnormalities (which were not compatible with reprogramming). We differentiated *TP53*^{mut} and *TP53*^{mut} del(5q) iPSCs into CD34⁺ HSPCs and established conditionally immortalized HSPCs, which can be continuously expanded for ~5 months while maintaining a developmental hierarchy with multipotent CD34⁺ HSPCs and more differentiated myeloid cell types.^{20,26} *TP53*^{mut} and *TP53*^{mut} del(5q) HSPCs expanded with similar cell division characteristics and maintained a stable karyotype during extended culture (Figure 1A; supplemental Figure 1A-B).

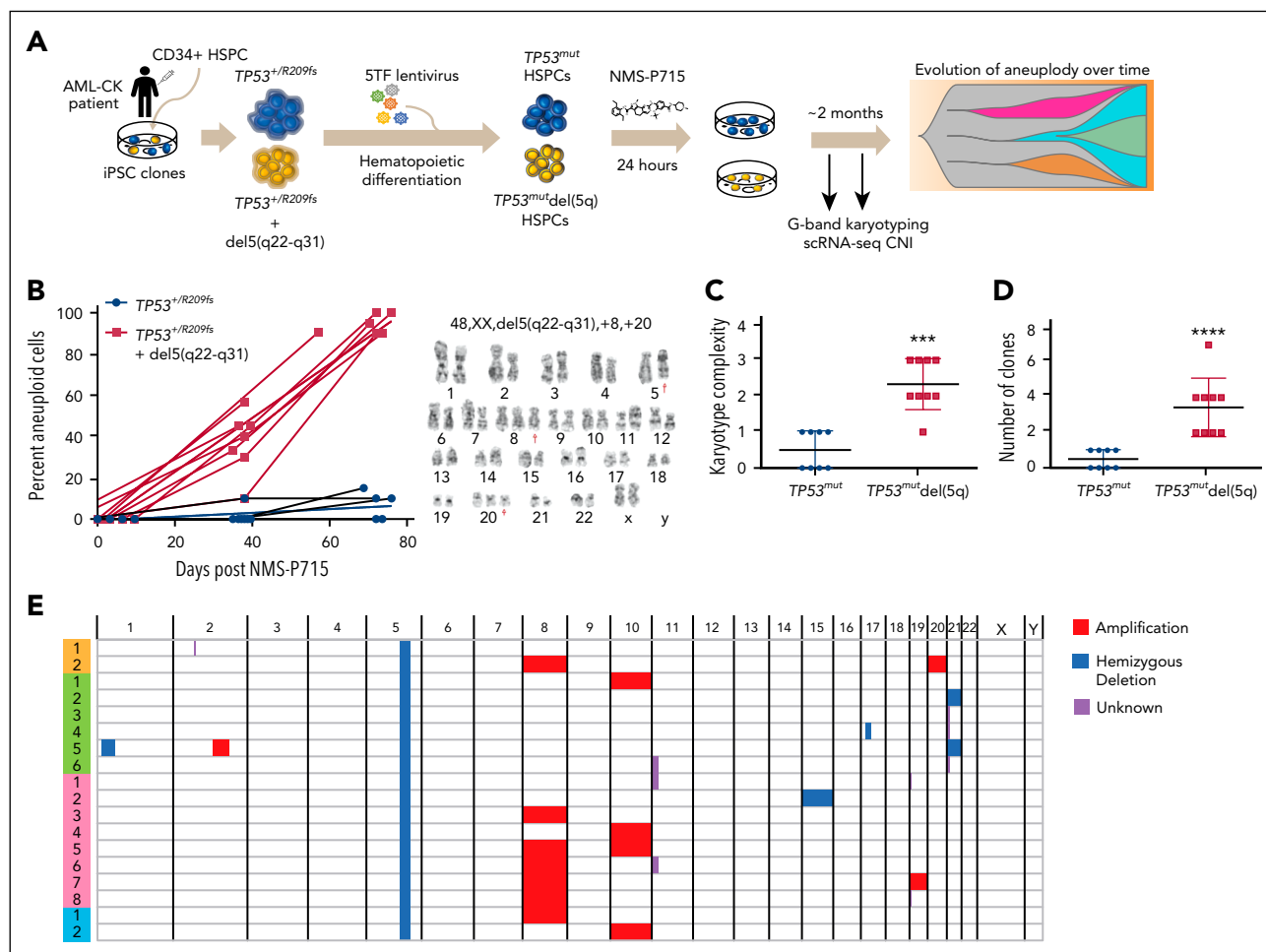


Figure 1. Experimental model of aneuploid evolution in $TP53$ -mutant preleukemic HSPCs. (A) Schematic showing reprogramming of $TP53^{mut}$ and $TP53^{mut} \text{ del}(5q)$ iPSCs from a patient with MDS/AML-CK and differentiation into iPSC-HSPCs. Cells were exposed to $1\mu\text{M}$ NMS-P715, an MPS1 inhibitor, for 24 hours, washed, and cultured continuously for 5 to 10 weeks. Cytogenetic analysis was performed using conventional karyotyping to analyze clonal dynamics of aneuploid cells. (B) Percentage of aneuploid cells over time after NMS-P715 treatment as reported by karyotyping (left), and a representative aneuploid clone with trisomy 8 and trisomy 20 evolved from $TP53^{mut} \text{ del}(5q)$ HSPCs (right). Connected line indicates measurements over time in the same culture ($n = 7$ independent cultures derived from $n = 3$ NMS-P715-treated samples). (C) Number of chromosome abnormalities per clone compared to parental lines (karyotype complexity); 8 independent cultures derived from 4 NMS-P715-treated samples, 2-tailed Mann-Whitney test ($***P = .002$). (D) Number of distinct aneuploid clones detected by karyotyping in each culture; $n = 8$ independent experiments; 2-tailed Mann-Whitney test ($****P < .0001$). (E) Summary of chromosome abnormalities identified by karyotyping in aneuploid clones derived from $TP53^{mut} \text{ del}(5q)$ HSPCs. Each row represents a separate clone, with experiments grouped by color ($n = 18$ clones derived from $n = 4$ NMS-P715-treated samples), individual chromosomes are in columns, and chromosome rearrangements are shown in color: red, amplification; blue, hemizygous deletion; purple, unknown material/rearrangement. Detailed information is included in Table 1. scRNAseq, single-cell RNA sequencing.

Errors in chromosome segregation because of defective mitotic checkpoint are a major mechanism of chromosome instability in cancer.^{3,30} MPS1 kinase inhibitors (MPSi) robustly induce chromosome missegregation, giving rise to aneuploid clones.^{31–33} We transiently treated $TP53^{mut}$ and $TP53^{mut} \text{ del}(5q)$ HSPCs with the MPSi NMS-P715 for 24 hours and monitored them by conventional karyotyping over an extended 5 to 10 weeks of culture (Figure 1A). Strikingly, we observed robust evolution to distinct highly aneuploid states after MPSi treatment across multiple experiments. Even more strikingly, $TP53^{mut} \text{ del}(5q)$ HSPCs almost invariably evolved to aneuploidy (mean, $\sim 95.1\%$), whereas parental $TP53^{mut}$ HSPCs infrequently gave rise to aneuploid clones (mean, 5.8%) ($P < .0001$; Figure 1B). $TP53^{mut} \text{ del}(5q)$ cultures harbored a greater number of abnormalities per clone ($P = .0002$) and distinct clones

($P < .0001$) than $TP53^{mut}$ HSPCs (Figure 1C-D). Clones derived from $TP53^{mut} \text{ del}(5q)$ HSPCs typically gained ~ 2 additional abnormalities, with several subclones observable by conventional karyotyping (at least 2/20 cells counted) after ~ 5 weeks of culture (Table 1). Abnormalities observed in the $TP53^{mut} \text{ del}(5q)$ MPSi culture were consistent with those of patients with AML-CK, including trisomy 8 (+8), trisomy 10 (+10), and unbalanced translocations or deletions of chromosome 21 (chr 21; Figure 1E; Table 1; supplemental Table 1). These observations contrasted with those seen in $TP53^{mut}$ HSPCs, which never reached $>20\%$ of total culture (Figure 1B). Overall, our findings show that iPSC-derived HSPCs undergo robust chromosome evolution after mitotic checkpoint inhibition and that 5q loss is a cooperating abnormality required for complex karyotype evolution.

Table 1. Karyotypes from HSPC aneuploid evolution experiments

Exp	Genotype	Sample	Treatment	Days	% Aneuploid	Karyotype
1	Pooled	1A	NMS	38	57	46,XX,del(5)(q22q31)[16]/46,idem,add(2)(p13)[16]/48,idem,+8,+20[5]/46,XX[3] (6/16 del5q nonclonal)
2	TP53	P53-2A	NMS	35	0	46,XX[30] (3 nonclonal)
2	TP53	P53-2A	NMS	76	10	47,XX,+8[2]/46,XX[18] (3/18 nonclonal)
2	del(5q)	del(5q)-2A	NMS	35	33	46,XX,del(5)(q22q31)[20]/47,sl,+10[4]/45,sl,-21[3]/46,sl,add(21)(p11.2)[3]
2	del(5q)	del(5q)-2A	NMS	76	100	46,XX,del(5)(q22q31),add(21)(p12)[12]/46,idem,add(11)(p15)[8]
2	TP53	P53-2A (rethaw)	NMS	57	0	46,XX[22] (4/22 nonclonal)
2	del(5q)	del(5q)-2A (rethaw)	NMS	57	100	45,XX,del(5)(q15q31),del(1)(p36.1p?32),dup(2)(q21q31),-21[18]/46,XX,del(5)(q15q31),add(21)(p11.2)[2]
3	TP53	P53-3 control	DMSO	60	15	46,XX,dup(1)(q21q32)[3]/46,XX[17] (2 nonclonal)*
3	TP53	P53-3A	NMS	38	10	46,XX,dup(1)(q21q32)[2]/48,XX,+10,+19[2]/46,XX[16] (2 nonclonal)*
3	TP53	P53-3B	NMS	38	0	46,XX,dup(1)(q21q32)[2]/46,XX[18] (2 nonclonal +3/+6)*
3	TP53	P53-3C	NMS	38	0	46,XX,dup(1)(q21q32)[3]/46,XX[17] (2 nonclonal -15/-17)*
3	TP53	P53-3D	NMS	38	0	46,XX,dup(1)(q21q32)[6]/46,XX[14] (3/14 nonclonal -8-13/-5-14-15/+1+7)*
3	TP53	P53-3A	NMS	72	10	46,XX,dup(1)(q21q32)[2]/46,XX,+del(1)(p13),-17[2]/46,XX[16]*
3	TP53	P53-3B	NMS	72	0	46,XX,dup(1)(q21q32)[3]/46,XX[17]*
3	TP53	P53-3C	NMS	72	15	46,XX,dup(1)(q21q32)[2]/46,XX,+1,add(1)(p12),del(1)(q12),-17[3]/46,XX[15]*
3	TP53	P53-3D	NMS	72	0	46,XX,dup(1)(q21q32)[4]/46,XX[16]*
3	del(5q)	del(5q)-3 control	DMSO	60	0	46,XX,del(5)(q22q33)[20]
3	del(5q)	del(5q)-3A	NMS	38	10	46,XX,del(5)(q22q31)[18]/45,XX,sl,-15[2] (4 nonclonal -19/-20/-11-18/+8-19)
3	del(5q)	del(5q)-3B	NMS	38	30	46,XX,del(5)(q22q31)[14]/45,sl,dic(11;19)(p15;q13.3)[3]/47,sl,+10[2]/47,sl,+8[1]
3	del(5q)	del(5q)-3C	NMS	38	40	46,XX,del(5)(q22q31)[12]/47,idem,+8[8] (3 nonclonal -10/-18/-21)
3	del(5q)	del(5q)-3D	NMS	38	45	46,XX,del(5)(q22q31)[11]/47,idem,+8[8]/47,idem,+10[1] (3 nonclonal -11/-13/-16)
3	del(5q)	del(5q)-3A+B+C+D	NMS	57	90	46,XX,del(5)(q22q33)[2]/47,XX,del(5),+8[15]/48,XX,del(5),+8,+10[4] (7 nonclonal in +8 samples)
3	del(5q)	del(5q)-3A	NMS	72	90	46,XX,del(5)(q22q33)[2]/47,XX,del(5)(q22q33),+8[12]/47,XX,del(5)(q22q33),+10[1]/47,XX,del(5)(q22q33),+8,add(11)(p15)[4]/47,XX,del(5),+10,add(11)(p15),add(19)(q13.4)[1]
3	del(5q)	del(5q)-3B	NMS	72	95	46,XX,del(5)(q22q33)[1]/47,XX,del(5)(q22q33),+8[6]/48,XX,del(5)(q22q33),+8,+10[2]/48,XX,del(5)(q22q33),+8,+19[6]/47,XX,del(5)(q22q33),+8,add(11)(p15)[2]/47,XX,del(5)(q22q33),+10,add(11)(p15)[1]/47,XX,del(5)(q22q33),+8,add(19)(q13.4)[2]/45,XX,del(5),+8,add(11)(p15),-19[1]

Complete karyotype reports from *TP53^{mut}* and *TP53^{mut} del(5q)* HSPC experiments after treatment with NMS-P715 or DMSO vehicle control shown in Figure 1B,E. Columns are organized as follows: experiment (1-4); genotype, in which TP53 indicates *TP53^{mut}* and del(5q) indicates *TP53^{mut} del(5q)*; sample name (A, B, C, etc); treatment (NMS or DMSO control); days, the culture period after NMS treatment at which karyotype was determined; and % aneupl, the percentage of cells with altered karyotype relative to the parental line. New abnormalities detected are shown in *italic*. Experiment 3 corresponds to the parallel evolution experiment in Figure 2D, in which subcultures A, B, C, and D were split from the same primary culture. In experiment 3, a spontaneous dup(1)(q21q32) abnormality was observed in *TP53^{mut}* DMSO control cultures, and clones with this abnormality (marked with *) were not included in the calculated percentage of aneuploid cells for NMS-treated cultures from the same original culture.

DMSO, dimethyl sulfoxide; Exp, experiment.

Table 1 (continued)

Exp	Genotype	Sample	Treatment	Days	% Aneuploid	Karyotype
3	del(5q)	del(5q)-3C	NMS	72	90	46,XX,del(5)(q22q33)[2]/47,XX,del(5)(q22q33),+8[13]/47,XX,del(5)(q22q33),+10[1]/48,XX,del(5)(q22q33),+8,+10[3]/47,XX,del(5)(q22q33),+8,add(11)(p15),add(19)(q13.4)[1]
3	del(5q)	del(5q)-3D	NMS	72	100	47,XX,del(5)(q22q33),+8[11]/47,XX,del(5)(q22q33),+10[4]/48,XX,del(5)(q22q33),+8,+10[3]/48,XX,del(5)(q22q33),+8, add(11)(p15)[2]
4	TP53	P53-4 control	DMSO	38	0	46,XX[20]
4	TP53	P53-4A	NMS	38	0	46,XX[20]
4	del(5q)	del(5q)-4 control	DMSO	38	15	46,XX,del(5)(q22q31)[17]/46,XX,del(5)(q22q31),dic(11;19)(p15;q13.3)[3]
4	del(5q)	del(5q)-4A	NMS	38	45	46,XX,del(5)(q22q31)[11]/47,XX,del(5)(q22q31),+8[7]/47,XX,del(5)(q22q31),+10[2]

Complete karyotype reports from *TP53^{mut}* and *TP53^{mut}* del(5q) HSPC experiments after treatment with NMS-P715 or DMSO vehicle control shown in Figure 1B,E. Columns are organized as follows: experiment (1-4); genotype, in which TP53 indicates *TP53^{mut}* and del(5q) indicates *TP53^{mut}* del(5q); sample name (A, B, C, etc); treatment (NMS or DMSO control); days, the culture period after NMS treatment at which karyotype was determined; and % aneupl, the percentage of cells with altered karyotype relative to the parental line. New abnormalities detected are shown in *italic*. Experiment 3 corresponds to the parallel evolution experiment in Figure 2D, in which subcultures A, B, C, and D were split from the same primary culture. In experiment 3, a spontaneous dup(1)(q21q32) abnormality was observed in *TP53^{mut}* DMSO control cultures, and clones with this abnormality (marked with *) were not included in the calculated percentage of aneuploid cells for NMS-treated cultures from the same original culture.

DMSO, dimethyl sulfoxide; Exp, experiment.

Evolution of aneuploidy is a stepwise process shaped by clonal competition

The clonal trajectories by which preleukemic *TP53*-mutant HSPCs evolve to aneuploidy in AML-CK are unknown. To examine these evolutionary dynamics, we followed paired *TP53^{mut}* del(5q) “evolving” and *TP53^{mut}* “nonevolving” HSPCs in long-term culture after a segregation error (Figure 2A). In a representative experiment, 3 independent *TP53^{mut}* del(5q) subclones emerged ~1 month after MPSi treatment with additional material on chr 21 [del(5q),add(21p)], monosomy 21 [del(5q),–21], and trisomy 10 [del(5q),+10] (Figure 2B top; supplemental Figure 2A left). After another month in culture, add(21p) emerged as the dominant clone, reaching fixation (~100%), whereas the other clones were no longer detectable by conventional karyotyping (Figure 2B top). Interestingly, the add(21p) clone further diversified by acquiring add(11p), which was not detected at the previous time point (Figure 2B; supplemental Figure 2A middle). The paired *TP53^{mut}* culture exhibited limited karyotype changes, with the presence of a minor +8 clone at ~10% (supplemental Figure 2B). Whole-genome sequencing showed that a limited number of single-nucleotide variants were acquired during aneuploid evolution, were private to individual lines, and did not map to known cancer driver genes (supplemental Table 2). These data suggest that chromosome evolution is driven by clonal competition, with a single clone emerging as the “winner” and further gradually diversifying.

The initial missegregation event is expected to produce small numbers of abnormal clones that may be lost by genetic drift. To test this possibility, we set up a parallel culture from a cryopreserved sample immediately after MPSi treatment (Figure 2A). By repeating the evolution experiment, we observed a completely different winning clone, with deletion 1p [del(5q),del(1p),dup(2),–21] gaining dominance over the

add(21p) clone, the latter failing to further diversify (Figure 2B bottom; supplemental Figure 2A right). One possible explanation is the bottleneck effect, wherein the rare del(1p) clone was randomly split into the cryopreserved sample. To determine if individual abnormalities act as drivers, we estimated the relative fitness of each clone using a Bayesian probabilistic framework designed to fit existing clone frequency measurements over time.³⁴ Notably, the add(21p) clone had a relative fitness advantage (mean 1 + s ~1.014) in the original culture, in which the del(1p) clone was not detected (Figure 2C top). By contrast, its fitness advantage was lost in the repeat culture (mean 1 + s ~0.971; Figure 2C bottom). These data suggest that certain chromosome abnormalities act as drivers to confer the highest relative fitness, with the repertoire of competing clones shaped by genetic drift and clonal selection.

To better examine evolutionary dynamics in early culture, we modeled a long-term evolution experiment^{35,36} by splitting an initial culture into 4 separate subcultures (A-D) after 24-hour MPSi treatment (Figure 2D). We observed multiple competing subclones in the 4 cultures, notably distinct from those seen in the previous experiment, followed by emergence of the +8 founder as the “winning” clone in all cultures (Figure 2E). Interestingly, the +8 clone further diversified by acquiring unique abnormalities in each culture, including add(11), +19, or +10 (Figure 2E). We observed some diversification of the founding clone in *TP53^{mut}* cultures, but with limited selection (supplemental Figure 2C). Mixing of the cultures after ~2 weeks resulted in a similar pattern of clonal expansion of the +8 founder (supplemental Figure 2D). Estimates of relative fitness (1 + s) varied across subcultures but were consistent with +8 acting as the driver event (Figure 2F). Overall, these data suggest that AML-CK evolution is shaped by genetic drift, which limits the initial repertoire of aberrant clones, with clonal

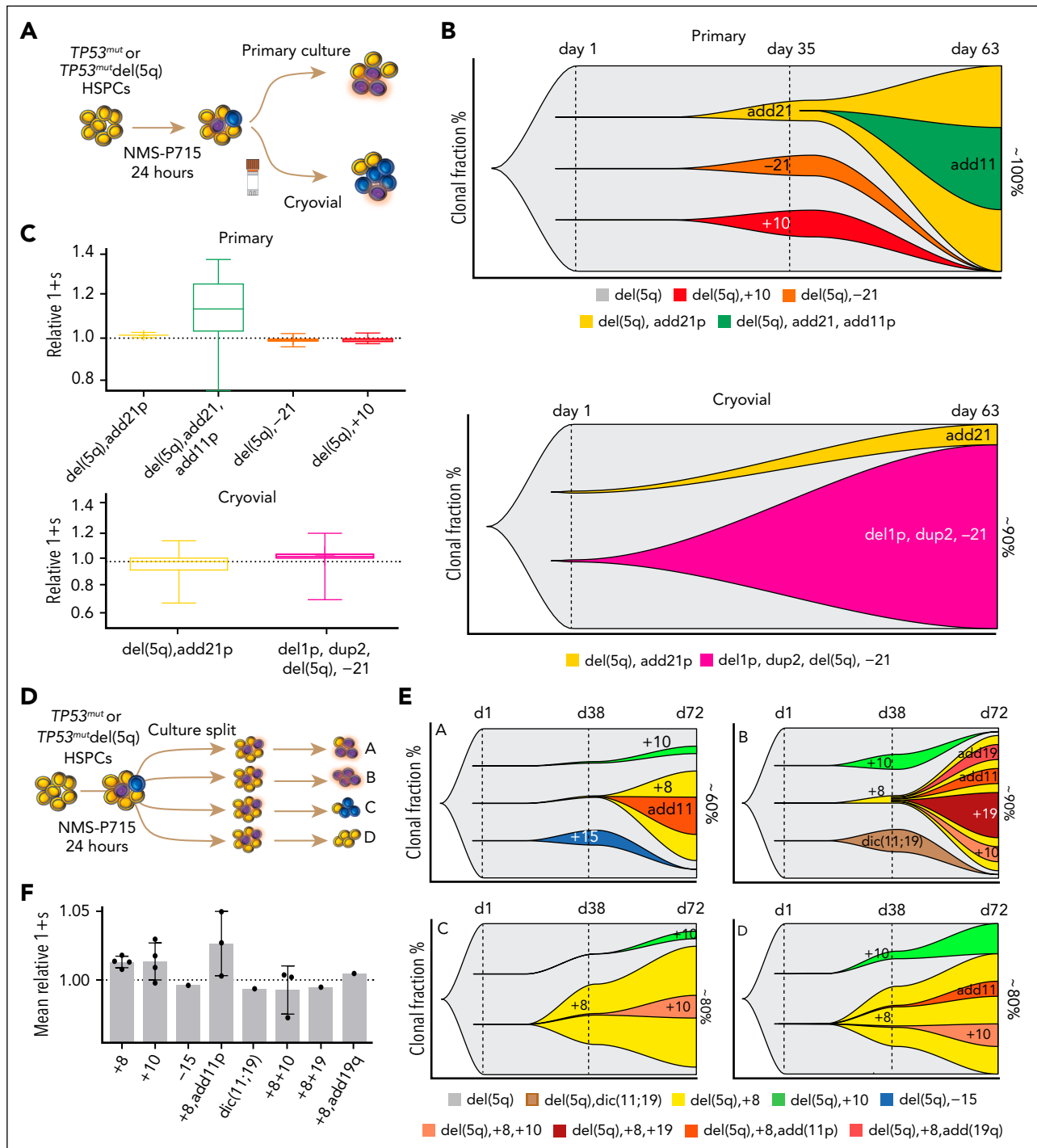


Figure 2. Evolution of aneuploidy is a stepwise process shaped by clonal competition. (A) Experimental scheme for HSPC evolution to aneuploidy after 24-hour treatment with the MPSi NMS-P715. A portion of the culture was cryopreserved, thawed, and compared to the primary culture. (B) Fish plot showing clonal dynamics in a *TP53^{mut}del(5q)* evolution experiment, with the primary experiment (top plot) and the repeat experiment from cryopreserved sample (bottom plot). The proportion of each clone over time is shown as determined by manual karyotyping at time points indicated by dashed lines. Individual clones are shown in colors, with inferred lineage indicated. (C) Mean relative fitness ($s + 1$) of each clone in the primary vs cryopreserved cultures in panel B; mean and 1st to 99th percentile of 9000 replicates. Dotted line indicates neutral fitness. (D) Experimental scheme of the parallel culture experiment, in which *TP53^{mut}* or *TP53^{mut}del(5q)* HSPCs were split into 4 separate cultures (A, B, C, D) immediately after 24-hour treatment with the MPSi NMS-P715, cultured in parallel, and analyzed by karyotyping at time points indicated by dashed lines. (E) Fish plot showing clonal dynamics in parallel cultures A, B, C, D, with the proportion of each clone determined by manual karyotyping. Clones are shown in colors, with inferred lineage indicated. (F) Mean relative fitness ($s + 1$) for each clone in parallel cultures in panel D. Neutral fitness ($s + 1 = 1$) is shown with dotted line. Each point is a mean of 9000 replicates for each culture; bars indicate mean $\pm$ standard deviation.

competition selecting a winning founder clone with the highest relative fitness, followed by subsequent evolution of progressively fitter clones. These findings help explain the highly

variable karyotypes of individual patients with AML-CK, which often contain derivatives of a single dominant clone (eg, a stemline).

Single-cell analysis identifies gene expression signature of aneuploidy

Conventional cytogenetic analysis is a gold standard in clinical practice but is limited to manual examination of a small number of metaphases. To gain deeper insight into clonal dynamics and concurrent cell state changes, we used single-cell RNA sequencing to quantify changes in gene expression in aneuploid clones identified by copy number inference (CNI).^{29,37} We profiled sequential samples in the representative experiment in Figure 2A by single-cell transcriptomics, followed by haplotype-aware copy number calling (Figure 3A; supplemental Figure 3A). As validation, unbiased calling of del(5q) in a combined *TP53*^{mut} and *TP53*^{mut} del(5q) data set revealed >95% confidence in correct calling (supplemental Figure 3B). Consistent with conventional karyotyping, the *TP53*^{mut} culture was comprised of several clones at <10% frequency, whereas the *TP53*^{mut} del(5q) culture reached ~60% abnormal cells (Figure 3B; supplemental Figure 3C). Unbiased lineage reconstruction further identified distinct normal and del(5q) clonal branches (Figure 3C). Analysis of *TP53*^{mut} del(5q) culture by conventional karyotyping showed the winning add(21p) clone (Figure 2B), but surprisingly, no chr 21 abnormalities were called by CNI. Instead, gain of chr 10q material (10q+) was identified (Figure 3C), suggesting that the additional material on chr 21p was derived from 10q. To confirm this, whole-genome sequencing identified a gain of 10q22qter, including the *PTEN* gene, which was localized to chr 21p by metaphase fluorescence in situ hybridization (supplemental Figure 3D-E). These data show that single-cell CNI enables robust identification of aneuploid clones, with improved ability to define the origin of rearranged material.

We next sought to define the gene expression changes associated with acquisition of aneuploidy in HSPCs. To this end, we compared gene expression between the ancestral del(5q) and the derived 10q+/add(21p) clone identified by CNI. Both cell types shared a similar distribution of myeloid, lymphoid, and progenitor cell populations (supplemental Figure 3F; supplemental Table 3). We identified 272 differentially expressed genes, including 116 upregulated genes, of which 37 were encoded within the 10q+ interval (supplemental Table 4).³⁸ Upregulated genes included the 10q *PTEN* tumor suppressor, which has an essential role in genome integrity,^{39,40} antiapoptotic factor *BCL2*, and *CD34* (Figure 3D). Upregulated genes were enriched for gene ontology terms related to mitosis and cohesin loading, the TGF β pathway, the p53 pathway, apoptotic signaling, and the cell cycle (Figure 3E). Key mitotic factors included cohesin subunits *STAG1*, *STAG2*, *WAPL*, and *SMC3*; regulators of sister chromatid segregation *CENPF*, *KIF11*, *KIF20B*, and *TOP2A*; and genome integrity factors *SHLD2*, *HELLS*, and *PTEN* (Figure 3D; supplemental Table 4), many of which are encoded in the 10q+ interval. These alterations suggest that aneuploid HSPCs upregulate chromosome segregation factors that may facilitate maintenance of the aneuploid cell state.

The aneuploidy signature is shared among patients with AML-CK

To determine if these gene expression changes are broadly associated with aneuploidy in hematopoietic cells, we next

performed single-cell transcriptomics on 2 primary samples from patients with AML-CK with *TP53* mutation and del(5q) (Figure 4A). CNI analysis of patient RO20518 identified del(5q) [add(5)(q11.2)], +8, chr 15 abnormalities [t(2;15)], and -16, found in the clinical karyotype (Figure 4B; supplemental Table 5). CNI analysis of patient R332 identified -5, -7, +8, +19, +21, and +22 by karyotype analysis (supplemental Figure 4A; supplemental Table 5) in addition to copy number changes that were not resolved by karyotype. The R332 sample comprised 4 highly complex CK subclones (CK1-4) (supplemental Figure 4A), whereas in the RO20518 sample, we also detected ancestral clones with isolated loss of 5q and 15p. To determine if the aneuploidy signature in our iPSC model was shared among patients with AML-CK, we generated a module score based on 116 genes upregulated in the +10q/add(21p) clone as the iPSC-CK signature (i-CK; supplemental Figure 4B). The i-CK signature was upregulated in all R332 CK1 to CK4 clones and in some RO20518 clones over normal copy number cells identified by CNI in each patient (Figure 4C). No chr 10q gain was detected by CNI or conventional karyotyping in either patient, suggesting that i-CK is a generalizable signature induced in aneuploid HSPCs in AML-CK irrespective of specific abnormalities. *PTEN* and the cohesin component *WAPL* were consistently upregulated in aneuploid clones in patients with AML-CK (Figure 4D-E). *BCL2* was upregulated in R332, but not RO20518, which expressed *CD14* (Figure 4D-E), consistent with the low expression of *BCL2* in myelomonocytic AML.⁴⁴ To characterize aneuploidy signature across larger patient cohorts, we analyzed single-cell RNA sequencing data from 5 patients with AML-CK reported by Wang et al,⁴¹ 8 patients with AML-CK reported by Leppä et al,¹⁴ and 1 patient with AML-CK from Petti et al.^{42,43} The samples included those with *TP53* mutation, *TP53*^{mut} with 17p/-17, and *TP53* wild-type cases, with multiple cytogenetic abnormalities detected by single-cell CNI (supplemental Figure 4C; supplemental Table 5). All 5 patients from Wang et al⁴¹ showed significant enrichment of the i-CK signature, including *PTEN* and cohesin, in complex subclones compared to euploid cells from the same patient (Figure 4F-G). Of the 8 patients from Leppä et al,¹⁴ 7 showed significant i-CK signature enrichment compared to normal immune cells from the same patient (supplemental Figure 4D). Similarly, i-CK signature was upregulated in the patient with AML-CK from Petti et al^{42,43} (supplemental Figure 5A-C), but not in patients with del(5q) MDS without CK (supplemental Figure 5D-F). Taken together, i-CK signature was broadly upregulated in 14 of 16 patients with AML-CK with diverse mutational status and cytogenetic abnormalities. i-CK signature was also largely distinct from leukemic stem cell (LSC-17) signature, which was not positively enriched in aneuploid clones (supplemental Figure 5G). We propose that a generalized gene expression state is associated with aneuploidy in AML-CK, despite different acquired chromosomal abnormalities.

BCL2 is a targetable molecular dependency of aneuploid clones

The antiapoptotic factor *BCL2*, upregulated in aneuploid HSPCs (Figure 3D), is the target of venetoclax, a *BCL2* inhibitor clinically used to treat high-risk AML.⁴⁵⁻⁴⁷ We sought to determine whether aneuploid clones are sensitized to venetoclax and how evolution to aneuploidy is affected by

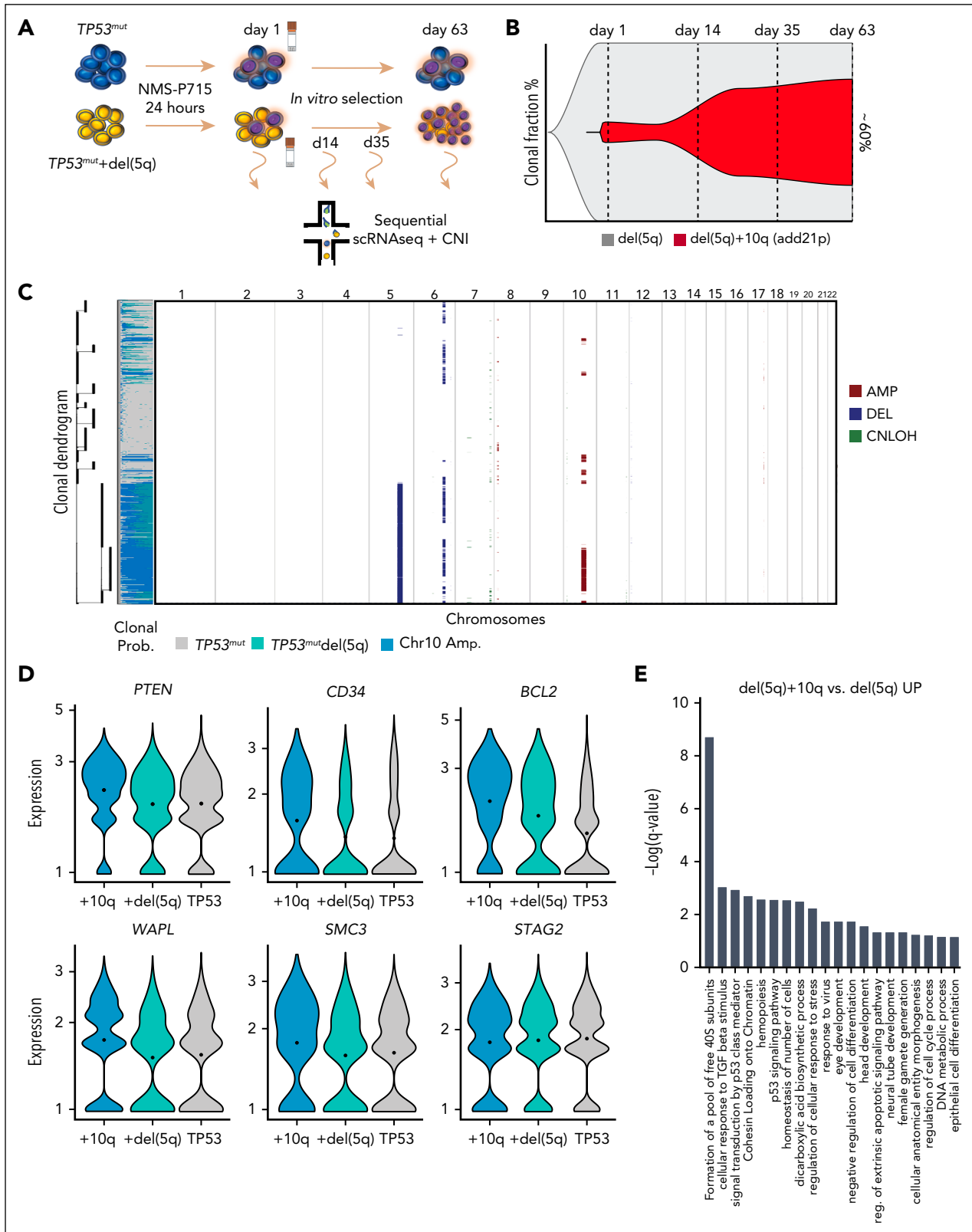


Figure 3. Single-cell analysis identifies a gene expression signature of aneuploidy. (A) Experimental scheme of the time-course experiment. *TP53^{mut}* and *TP53^{mut} del(5q)* HSPCs were evolved and sequentially sampled by scRNAseq at the indicated time points, and clonal makeup was defined by CNI. (B) Fish plot showing clonal dynamics of the *TP53^{mut} del(5q)* culture corresponding to Figure 2B, as determined by single-cell CNI using Numbat at time points indicated by dashed lines. (C) Copy number abnormalities in single cells identified using Numbat; data combined from all time points in *TP53^{mut}* and *TP53^{mut} del(5q)* lines. CNAs are shown in color: red, amplification; blue, deletion; green, copy number–neutral loss of heterozygosity. Clonal identity, based on the probability that the cell belongs to each inferred clone, is shown as the color bar on the left: gray, *TP53^{mut}* only; green, *TP53^{mut} del(5q)*; blue, *TP53^{mut} del(5q)+10q*. (D) Normalized gene expression level in single-cell clones identified by Numbat; log scale; black dots indicate mean. (E) Top gene ontology categories in Metascape for genes upregulated in aneuploid *TP53^{mut} del(5q)+10q* compared to parental *TP53^{mut} del(5q)* HSPCs. Upregulated genes were defined by Wilcoxon rank-sum test, $q < 0.05$. $-\log(q\text{-value})$ for each term is displayed ($q < 0.05$). scRNAseq, single-cell RNA sequencing; TGF, transforming growth factor.

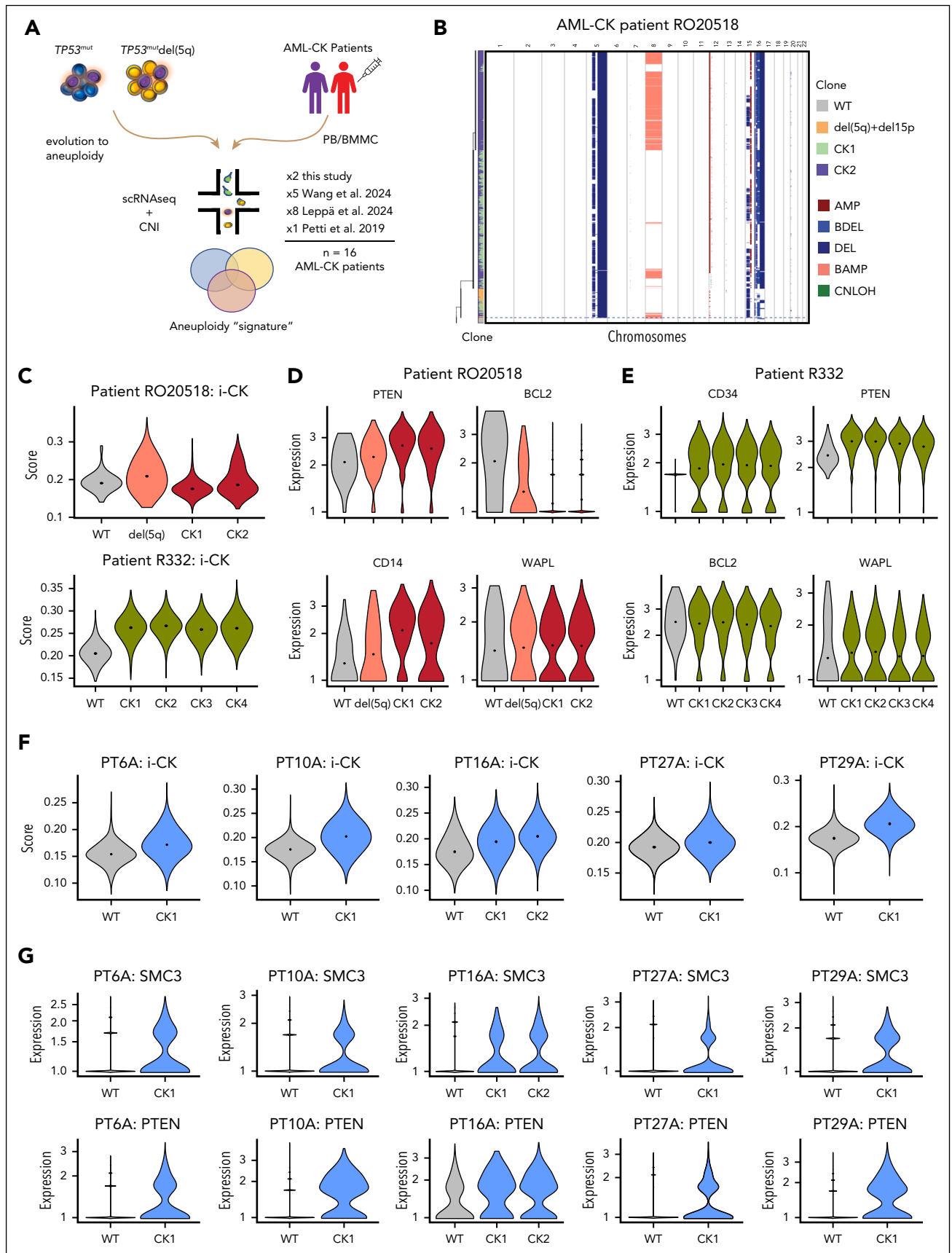


Figure 4. The aneuploidy signature is shared among patients with AML-CK. (A) Experimental scheme of single-cell transcriptomic analysis comparing $TP53^{mut}$ del(5q) iPSC-HSPCs and patients with AML-CK sequenced in this study, as well as published data sets from Leppä et al.,¹⁴ Wang et al.,⁴¹ and Petti et al.^{42,43} (B) Copy number

venetoclax treatment. Venetoclax response can be modulated by levels of BCL2, other antiapoptotic BCL2 family proteins, and proapoptotic factors (such as BAX).^{47,48} *TP53*^{mut} del(5q) and aneuploid +10q/add(21p) HSPCs selectively upregulated *BCL2* (Figure 3D), but not other antiapoptotic paralogs, such as *MCL1*, *BCL2L1*, or *BCL2A1* (Figure 5A). Consistent with this, parental *TP53*^{mut} del(5q) HSPCs were highly sensitive to venetoclax compared to *TP53*^{mut} HSPCs (50% inhibitory concentration [IC₅₀], ~0.250 μM vs ~40 μM; Figure 5B). To test how treatment with venetoclax affects evolution to aneuploidy, we evolved cryopreserved *TP53*^{mut} del(5q) HSPCs (Figure 2A) with continuous treatment with vehicle control or venetoclax at the 250 nM IC₅₀ dose (Figure 5C). In the vehicle-treated culture, the same add(21p) [del(5q),add(21p),add(11p)] clone gained relative dominance as in the original experiment, although, interestingly, it underwent a whole-genome duplication event to give rise to a 4N subclone (Figure 5D left). In marked contrast, the *BCL2*-high add(21p) clone was eradicated in the venetoclax-treated culture, and instead, the del(1p) [del(1p),-dup2,del(5q),-21] minor clone became dominant (Figure 5D right). The del(1p) founder clone massively diversified under venetoclax selection into several complex subclones, which gained dic(11;19) and -16 (Figure 5D right). The venetoclax-treated culture displayed a marked resistance to venetoclax (IC₅₀, ~70 μM) compared to the vehicle-treated culture with dominant add(21p) clone, which had even greater sensitivity to venetoclax (IC₅₀, ~0.02 μM) than parental *TP53*^{mut} del(5q) HSPCs (IC₅₀, 0.25 μM; Figure 5E). These data show that some aneuploid clones are sensitized to venetoclax, with therapy resistance driven by subclonal diversity.

Venetoclax-resistant aneuploid HSPCs undergo a lineage switch

To understand the mechanisms driving venetoclax resistance in AML-CK, we next performed single-cell karyotyping. This analysis readily detected the add(21p)/10q+ and the del(1p) clones harboring dup2 and -21 (Figure 5F). Venetoclax almost entirely eradicated add(21p)/10q+ and parental *TP53*^{mut} del(5q) clones, with the culture largely comprising the del(1p) clone (Figure 5G; supplemental Figure 6A-B). Interestingly, the venetoclax-resistant culture markedly downregulated *BCL2* expression and upregulated *BCL2A1* expression by quantitative polymerase chain reaction (Figure 6A). Single-cell analysis showed that *BCL2* expression was reduced in the del(1p) clone in both vehicle- and venetoclax-treated cultures (Figure 6B), suggesting that this shift is an intrinsic property of this aneuploid genotype rather than a response to venetoclax. Since HSPC populations express different levels of BCL2 factors, we next investigated HSPC composition. The add(21p) clone was largely similar to immature lymphoid-primed multipotent

progenitors. By contrast, the del(1p) clone showed a significant contribution from more mature erythroid-megakaryocyte progenitors (MEP-like) (Figure 6C; supplemental Table 6). Megakaryocyte and erythroid genes were induced in the gene ontology annotation of del(1p)-upregulated genes (Figure 6D), including *FLI1*, *TAL1*, and *ITGA2B* (CD41) (supplemental Figure 6C), and, interestingly, JAK/STAT genes *STAT1*, *STAT3*, and *STAT5B* (Figure 6E; supplemental Table 7). Importantly, the gene signature from the MEP-like del(1p) clone was enriched in a subset of patients with AML-CK, for example, RO20518, that expressed low *BCL2* and was not highly enriched for the i-CK signature (supplemental Figure 6D,E). The patient cohort from Leppä et al¹⁴ included a patient with AML-CK (P9) treated with venetoclax. The diagnostic sample comprised a dominant CK3 clone that expressed high i-CK signature/*BCL2* and immature markers (CD34, CD117) and a minor CK1 clone that expressed low i-CK signature/*BCL2* and MEP markers (CD71, CD41) (supplemental Figure 6F-I). Consistent with our model, venetoclax treatment depleted the immature CK3 clone, leading to clonal dominance of the MEP-like CK1 clone (supplemental Figure 6H). These data suggest that venetoclax-resistant aneuploid clones can undergo a lineage switch from more primitive to MEP-like progenitor cell identity that intrinsically expresses lower levels of *BCL2*.

Because aneuploid MEP-like HSPCs may be sensitized to distinct drug classes, we screened parental, vehicle-treated, and venetoclax-treated lines with a panel of US Food and Drug Administration–approved drugs for hematologic malignancies (supplemental Figure 7A; supplemental Table 8). Interestingly, venetoclax-resistant HSPCs with dominant del(1p) clones were sensitized to multiple JAK2 inhibitors clinically used to treat myeloproliferative neoplasms, including ruxolitinib, momelotinib, fedratinib, and pacritinib (supplemental Figure 7B). We further validated that venetoclax-resistant HSPCs were sensitized to ruxolitinib compared to vehicle-treated cells (Figure 6F). To test the combination of venetoclax and a JAK2 inhibitor, we treated aneuploid HSPCs with venetoclax as before (Figure 5C), allowing outgrowth of resistant clones, followed by treatment with ruxolitinib. Ruxolitinib at the IC₅₀ dose of 50 nM suppressed the outgrowth of the venetoclax-resistant MEP-like del(1p) clone by ~2.5-fold (Figure 6G-I). Lastly, we tested venetoclax and JAK2 inhibitors in 3 primary AML-CK samples. Two of these samples had a poor response to venetoclax alone (IC₅₀, >2 μM; supplemental Figure 7C) but were resensitized in combination with a JAK2 inhibitor, especially pacritinib (supplemental Figure 7D). Overall, these findings show that venetoclax monotherapy resistance can arise from aneuploid HSPCs with MEP-like lineage identity that are sensitized to distinct drug classes.

Figure 4 (continued) abnormalities identified in single cells of patient RO20518. Red, amplification; blue, deletion; green, copy number–neutral loss of heterozygosity. Clonal identity, based on the probability that the cell belongs to each inferred clone, is shown as the color bar on the left: gray, WT; orange, del(5q)+del(15p); green/blue, CK clones 1 to 2. (C) Normalized expression of the i-CK signature genes in single-cell clones of patients RO20518 and R332; log scale; the black dot indicates median. Statistical testing: in R332, all CK clones were enriched for i-CK ($P < .0001$) and RO20518 del(5q) ($P < .0001$) (Holm-adjusted Wilcoxon rank-sum test). (D) Normalized gene expression level in single-cell clones (WT, del(5q), CK1-2) in patient RO20518; log scale; the black dot indicates median. (E) Normalized gene expression level in single-cell clones (WT, CK1-4) in patient R332; log scale; the black dot indicates mean. (F) Normalized expression of the i-CK signature genes in single-cell aneuploid (CK) or euploid (WT) clones of patients with AML-CK from Wang et al⁴¹; the black dot indicates mean. Statistical testing: in PT6A, PT10A, PT16A, PT27A, and PT29A, all CK clones were enriched for i-CK ($P < .0001$; Holm-adjusted Wilcoxon rank-sum test). (G) Normalized gene expression level of *PTEN* and *SMC3* in single-cell euploid (WT) or aneuploid (CK) clones in patients with AML-CK from Wang et al⁴¹; log scale. BMMC, bone marrow mononuclear cells; PB, peripheral blood; WT, wild-type.

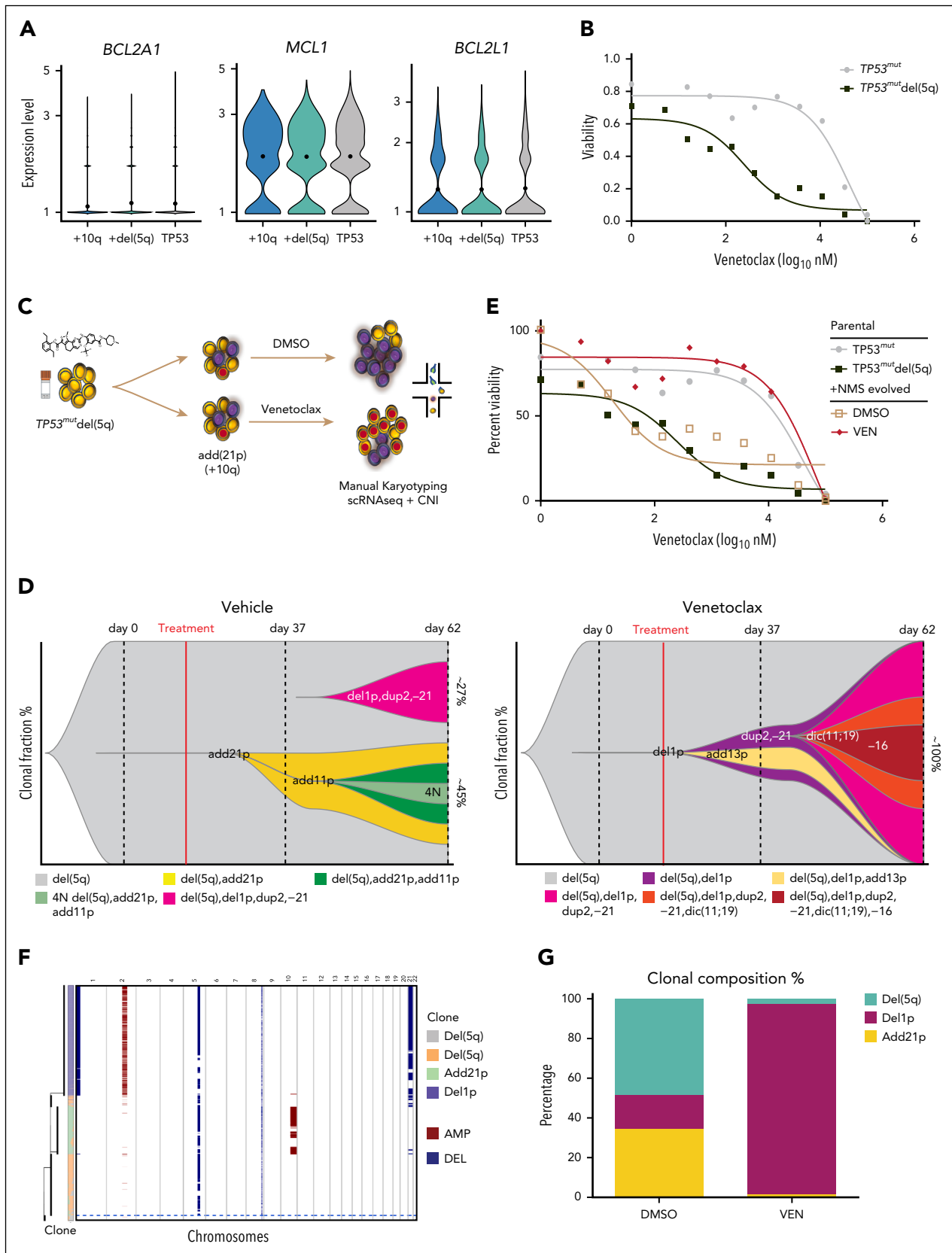


Figure 5. BCL2 is a targetable molecular dependency of aneuploid clones. (A) Normalized gene expression level in single-cell clones identified by Numbat from iPSC-HSPCs; log scale; the black dot indicates mean. (B) Relative cell viability of parental $TP53^{mut}$ and $TP53^{mut}del(5q)$ HSPCs treated with VEN in dose-response format for 7 days; also shown in panel E. Concentration is plotted as $\log_{10}$ scale (nanomolar). (C) Experimental scheme of VEN treatment, in which HSPCs harboring the *BCL2*-high $del(5q)+10q$ subclone (Figure 2B) were evolved with 250nM VEN or DMSO vehicle for ~48 days and sampled periodically for karyotype. (D) Fish plots showing clonal dynamics

Discussion

The clonal origins of myeloid neoplasms with complex karyotype remain largely unknown. Here, we developed an iPSC model system in which preleukemic HSPCs evolve to complex and diverse aneuploid states after transient mitotic inhibition. Using this model, we demonstrate that iPSC-derived AML-CK preleukemic HSPCs with *TP53* mutation and del(5q), but not *TP53* mutation alone, undergo complex karyotype evolution marked by sequential acquisition of chromosomal abnormalities. By tracking this process at single-cell resolution, we identified a gene expression signature shared among patients with AML-CK. We propose that despite distinct collections of chromosomal abnormalities, aneuploidy in AML-CK is associated with generalized gene expression states, reflecting the clonal origin and molecular dependencies of aneuploidy, such as antiapoptotic *BCL2* factors. Aneuploid clones were sensitized to standard-of-care venetoclax therapy, with resistance driven by lineage switching to HSPC states with low *BCL2* expression. These findings show the existence of molecular dependences in aneuploid HSPCs that can be therapeutically targeted in *TP53*-mutant leukemias.

AML-CK presents with widespread aneuploidy at diagnosis, making it difficult to reconstruct clonal origins. Near-complete penetrance of *TP53* mutations and their detection in clonal hematopoiesis and in preleukemic clones from patients with AML-CK^{10,11} suggest that these are initiating mutations. Most patients with AML-CK harbor recurrent del(5q), suggesting that this region encodes critical genes that cooperate with mutant *TP53* to drive AML-CK progression.^{7,17-19} Here, to our knowledge, we provide the first experimental evidence that 5q loss is a driver of AML-CK progression. Loss of 5q is also an early event in myeloid oncogenesis, and isolated del(5q) defines a subtype of MDS with favorable prognosis and macrocytic anemia,¹⁹ which can progress to high-risk MDS/AML via secondary mutations, including in *TP53*.⁴⁹ This suggests that del(5q) may be the initiating lesion in some patients, followed by secondary *TP53* mutations. The commonly deleted 5q region in AML-CK spans 5q22 to 5q31 and encodes multiple genes implicated in disease pathogenesis, including *EGR1*, *APC*, *CDC25C*, *CTNNA1*, and *LMNB1*.²⁰⁻²³ Identification of 5q tumor suppressor genes with roles in regulating genome stability remains an important goal. We propose a common origin of AML-CK that requires cooperative genetic loss of *TP53* and del(5q) and inhibition of mitotic checkpoint integrity via genotoxic or cytotoxic therapy.

Clonal trajectories of chromosomal evolution in the human hematopoietic system have not been studied experimentally. Chromosomal shattering, termed chromothripsis, is observed in genetically unstable tumors, which suggests rapid, catastrophic evolution.^{3,50} Evolution in our model is consistent with a more gradualist model of neoplastic progression, in which rare

aneuploid cells arising after mitotic checkpoint inhibition diversify in a sequential fashion. Other studies of *TP53*-mutated secondary AML are also suggestive of a sequential model of aneuploidy.⁵¹ Recent single-cell analyses in patients with AML-CK have uncovered evidence for both sequential and punctuated evolution,¹⁴ although detection of early clones remains rare.

By using single-cell CNI to track this evolutionary process, we identified an i-CK gene expression signature shared among patients with AML-CK. This signature is marked by upregulation of HSC genes and genome integrity factors, including *PTEN*⁴⁰ and cohesin subunits, including *SMC3* and *WAPL*. Aneuploidy imposes unique cellular stresses on proteostasis, replication, and mitosis.²⁴ Inappropriate chromatid pairing during mitosis induces high levels of genome instability that can be incompatible with cell survival.⁵² Interestingly, *PTEN* has a noncanonical role in promoting genome integrity in mitosis.³⁹ Cohesins mediate chromosome pairing in early mitosis, and cohesin cleavage is required for anaphase progression. Critically, although cohesin mutations in *STAG2* portend a poor prognosis in myeloid neoplasms, they are mutually exclusive with *TP53* mutations and are almost never observed in AML-CK.⁵³ We hypothesize that upregulation of *PTEN* and cohesins enables aneuploid cells to stabilize chromosome pairing in mitosis, explaining why diverse CK genotypes converge on this i-CK signature. Our model of AML-CK will enable further dissection of the mechanisms required for the maintenance of an aneuploid cell state in HSPCs.

TP53 mutation and complex karyotype define a high-risk subgroup of myeloid neoplasms with poor response to antileukemic therapies,^{17,25} including the *BCL2* inhibitor venetoclax.^{45,54-56} We find that *BCL2* is often upregulated as part of the i-CK signature and may promote the survival of aneuploid clones. Treatment with venetoclax eradicates clones expressing high *BCL2* levels; however, resistance is driven by aneuploid clones with reduced expression of *BCL2* and induction of alternative *BCL2* family factors, such as *BCL2L1* and *BCL2A1*. The switch to alternative *BCL2* factors characterizes normal hematopoietic differentiation; *BCL2* expression declines during myeloid differentiation with dependence on *MCL1*,⁴⁴ whereas erythroid and megakaryocyte lineages are dependent on *BCL2L1*.⁵⁷ Interestingly, AMLs in which LSCs display expression of mature monocytic or erythroid/megakaryocyte programs display poor responses to venetoclax.^{44,48,57} Consistent with this, our data suggest that venetoclax-resistant aneuploid clones undergo an erythroid/megakaryocyte lineage switch that sensitizes them to JAK2 inhibition. We propose that, similarly to nonaneuploid leukemias, LSCs in AML-CK likely exist in a spectrum of cell states, from more primitive (HSC-like) to more mature (MEP-like), among diverse clones with distinct molecular vulnerabilities.⁵⁸ More broadly, we propose that aneuploid cell states confer

Figure 5 (continued) of the *TP53*^{mut} del(5q) cultures evolved in the presence of DMSO vehicle (left) or VEN (right) from days 14 to 62, with the proportion of each clone determined by manual karyotyping. Clones are shown in colors, with inferred lineage indicated. (E) Relative cell viability of *TP53*^{mut} del(5q)-derived cultures evolved with VEN or DMSO and parental lines (also shown in panel B) treated with VEN in dose-response format for 7 days. Concentration is plotted as log₁₀ scale (nanomolar). (F) Copy number abnormalities identified by Numbat in single cells of pooled vehicle and VEN-treated samples. Red, amplification; blue, deletion. Clonal identity, based on the probability that the cell belongs to each inferred clone, is shown as the color bar on the left: gray/orange, del(5q) parental; green, add21p/+10q; blue, del(1p). (G) Percent clonal contribution for each culture as determined by Numbat single-cell CNI; add21p, [del(5q);add(21p)/+10q]; del1p, [del(5q);del(1p);dup(2);-21]. DMSO, dimethyl sulfoxide; VEN, venetoclax.

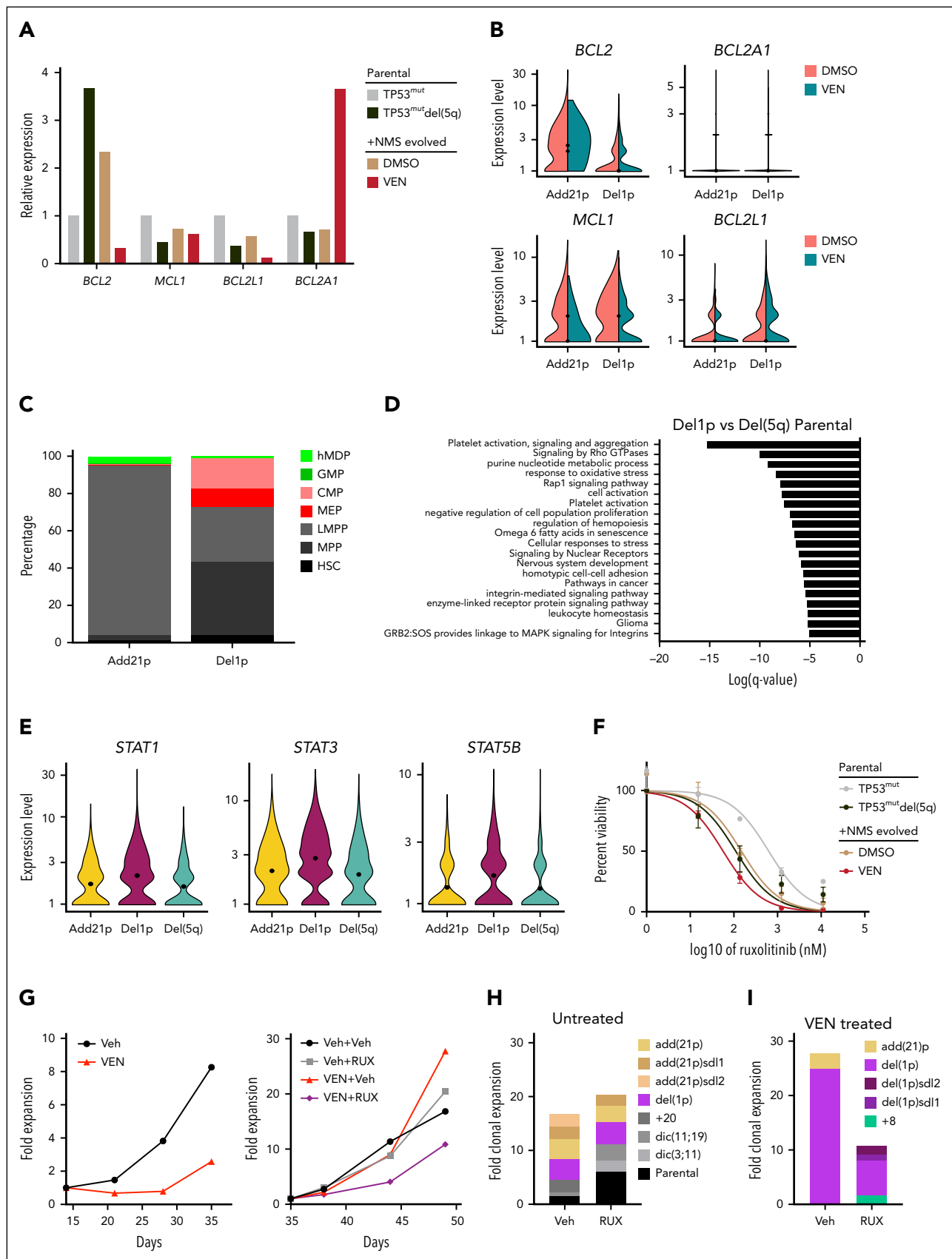


Figure 6. VEN-resistant aneuploid HSPCs undergo a lineage switch. (A) Expression of BCL2 family genes in bulk $TP53^{mut}$ del(5q) HSPCs evolved in the presence of vehicle or VEN compared to parental lines. Expression was measured relative to *GAPDH* by quantitative polymerase chain reaction and normalized to $TP53^{mut}$ HSPCs. (B) Violin plots of expression of BCL2 family genes in single cells with clonal identity determined by CNI. Color indicates treatment condition (vehicle or VEN); log scale; the

Figure 6 (continued) black dot indicates mean. (C) Lineage composition identified by SingleR for add(21p) and del(1p) clones determined by CNI. Cell types with <5 cells in both conditions were excluded. (D) Top gene ontology categories in Metascape analysis of 284 upregulated genes (log2 fold change > 0.5; Wilcoxon rank-sum test $q > 0.05$) in the del(1p) clone vs the parental *TP53*^{mut} del(5q) clone. Log(q value) of each term is displayed ($q < 0.05$). (E) Violin plots of expression of STAT family genes differentially expressed between del(1p) and *TP53*^{mut} del(5q) parental HSPCs, colored by clone; log scale; the black dot indicates mean. (F) Normalized relative mean cell viability of aneuploid cultures treated with the JAK2 inhibitor RUX. *TP53*^{mut} del(5q) HSPCs evolved in the presence of vehicle or VEN, and parental lines were treated with RUX in dose-response format for 5 days. $n = 2$; concentration is plotted as log₁₀ scale (nanomolar). (G) Cell growth of HSPCs from the add(21p)/del(1p) culture (Figure 5C) treated sequentially with VEN and the JAK2 inhibitor RUX. HSPCs were treated with 250nM VEN or DMSO vehicle from days 14 to 35, and subsequently treated with 50nM RUX or vehicle from days 35 to 49. Cell growth of vehicle- or VEN-treated cultures as fold expansion relative to day 14 (left). Cell growth of vehicle- or RUX-treated cultures as fold expansion relative to day 35 (right). (H) Clonal expansion of VEN-untreated cultures treated with vehicle or RUX. Clonal composition was determined by karyotyping, with fold expansion relative to day 35. (I) Clonal expansion of VEN-treated cultures subsequently treated with vehicle or RUX. Clonal composition was determined by karyotyping, with fold expansion relative to day 35. CMP, common myeloid progenitor; GMP, granulocyte-monocyte progenitor; hMDP, human monocyte-dendritic cell progenitor; LMPP, lymphomyeloid primed progenitor; MEP, megakaryocyte-erythroid progenitor; MPP, multipotent progenitor; RUX, ruxolitinib; Veh, vehicle.

unique molecular vulnerabilities that can be therapeutically targeted in high-risk, *TP53*-mutant leukemias.

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Authorship

Contribution: J.P.C. and S.D. conceived the project and designed the experiments; J.P.C. performed the experiments with assistance from S.R., S.S., V.J., and J.W.; J.P.C., S.R., S.G., F.Y.H., A.-M.L., V.J., P.S.B., A.T., M.F., E.M.S., and S.D. analyzed and interpreted the data; J.L.A., J.S.A., S.H.K., P.S.B., A.T., V.J., M.F., and E.M.S. provided input

on experimental design and interpretation; B.W. and H.A.A. provided patient sample data; A.N.S., S.H.K., J.S.A., J.L.A., and E.M.S. provided patient samples and clinical interpretation; and J.P.C. and S.D. prepared the manuscript.

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Footnotes

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Raw and processed single-cell RNA sequencing data and copy number inference analysis have been deposited in the Gene Expression Omnibus (accession number GSE282370).

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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